

SET 3: EMPLOYEE PERFORMANCE EVALUATION

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Dataset: Employee Performance

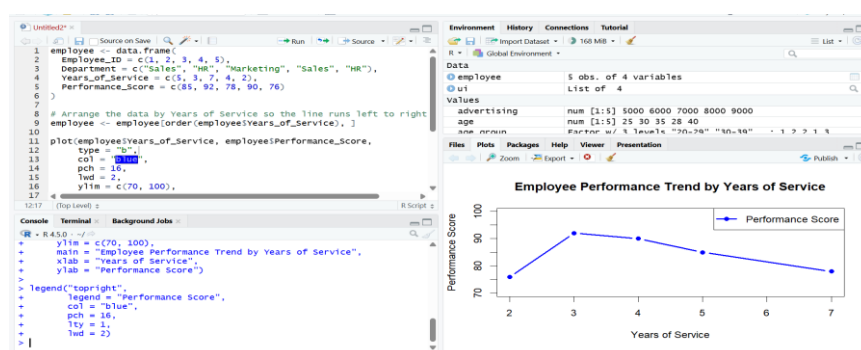
Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

PROGRAM 1: Line Chart of Employee Performance

Question: Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

Code:

```
employee <- data.frame(  
  Employee_ID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(85, 92, 78, 90, 76)  
)  
  
# Arrange the data by Years of Service so the line runs left to right  
employee <- employee[order(employee$Years_of_Service), ]  
  
plot(employee$Years_of_Service, employee$Performance_Score,  
  type = "b",  
  col = "blue",  
  pch = 16,  
  lwd = 2,  
  ylim = c(70, 100),  
  main = "Employee Performance Trend by Years of Service",  
  xlab = "Years of Service",  
  ylab = "Performance Score")  
  
legend("topright",  
  legend = "Performance Score",  
  col = "blue",  
  pch = 16,  
  lty = 1,  
  lwd = 2)
```



Output 1: Employee Performance Trend Line Chart

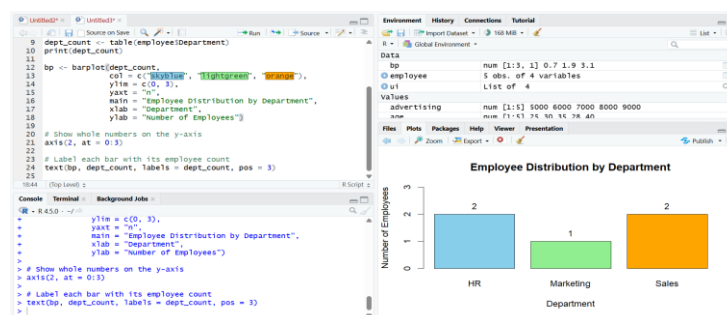
Observation: The line chart shows the performance score of the employees against their years of service. The score is lowest (76) at 2 years of service and highest (92) at 3 years. After 3 years, the score falls steadily to 90 at 4 years, 85 at 5 years and 78 at 7 years. So the trend rises at first and then declines as years of service increase in this dataset.

PROGRAM 2: Bar Chart of Employees by Department

Question: Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Code:

```
employee <- data.frame(  
  Employee_ID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(85, 92, 78, 90, 76)  
)  
  
# Count the number of employees in each department  
dept_count <- table(employee$Department)  
print(dept_count)  
  
bp <- barplot(dept_count,  
              col = c("skyblue", "lightgreen", "orange"),  
              ylim = c(0, 3),  
              yaxt = "n",  
              main = "Employee Distribution by Department",  
              xlab = "Department",  
              ylab = "Number of Employees")  
  
# Show whole numbers on the y-axis  
axis(2, at = 0:3)  
  
# Label each bar with its employee count  
text(bp, dept_count, labels = dept_count, pos = 3)
```



Output 2: Employee Distribution by Department

HR	Marketing	Sales
2	1	2

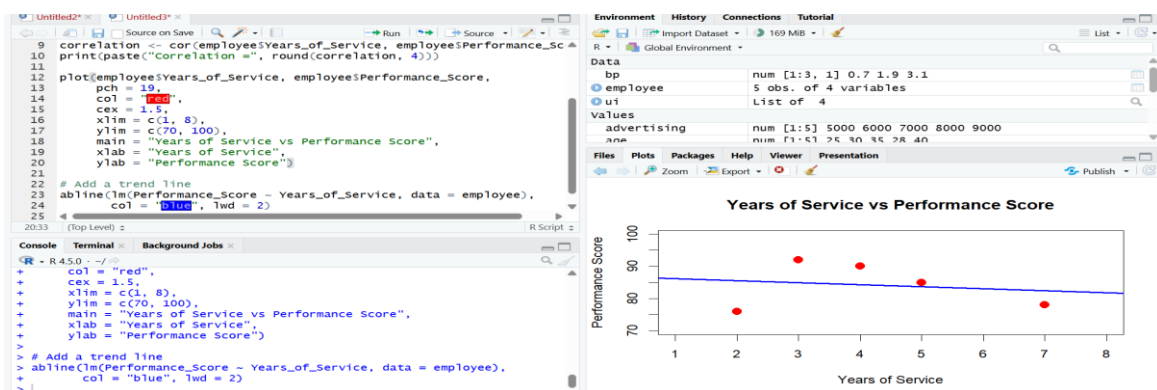
Observation: The bar chart shows that HR and Sales each have 2 employees, while Marketing has only 1 employee. Out of the 5 employees in the dataset, HR and Sales together account for 4, so Marketing is the smallest department.

PROGRAM 3: Scatter Plot of Years of Service vs Performance Score

Question: Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

Code:

```
employee <- data.frame(  
  Employee_ID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(85, 92, 78, 90, 76)  
)  
  
# Calculate the correlation coefficient  
correlation <- cor(employee$Years_of_Service, employee$Performance_Score)  
print(paste("Correlation =", round(correlation, 4)))  
  
plot(employee$Years_of_Service, employee$Performance_Score,  
  pch = 19,  
  col = "red",  
  cex = 1.5,  
  xlim = c(1, 8),  
  ylim = c(70, 100),  
  main = "Years of Service vs Performance Score",  
  xlab = "Years of Service",  
  ylab = "Performance Score")  
  
# Add a trend line  
abline(lm(Performance_Score ~ Years_of_Service, data = employee),  
  col = "blue", lwd = 2)
```



Output 3: Years of Service vs Performance Score Scatter Plot

```
[1] "Correlation = -0.1688"
```

Observation: The correlation between years of service and performance score is -0.1688 (about -0.17). The negative sign means that, in this dataset, employees with more years of service tend to have slightly lower performance scores. Since the value is close to 0, the correlation is weak. The points are scattered widely around the trend line, so there is no clear

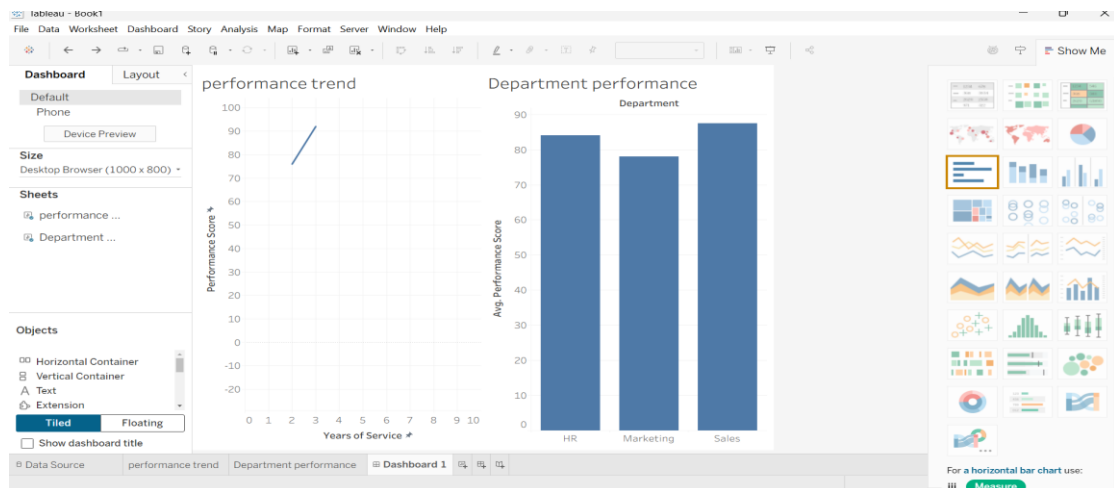
linear pattern. This result only describes the observed relationship in these 5 records; it does not show that years of service causes performance to increase or decrease, and a larger dataset would be needed for a reliable conclusion.

PROGRAM 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Steps to create the dashboard in Tableau:

1. Save the Employee Performance dataset (Employee ID, Department, Years of Service, Performance Score) as an Excel or CSV file.
2. Open Tableau, click Connect to a File (Microsoft Excel or Text file), select the file and open the Sheet 1 tab.
3. Right-click Employee ID and Years of Service in the Data pane and choose Convert to Dimension, so they are not summed as measures.
4. Line chart: rename the sheet as "Performance Trend". Drag Years of Service to Columns and Performance Score to Rows. Change the Marks type to Line, add Circle marks or enable Show Mark Labels, and drag Department to Color to show the legend. Give the sheet a suitable title and axis labels.
5. Bar chart: open a new sheet and rename it as "Department Distribution". Drag Department to Columns and Employee ID to Rows, then change the aggregation to Count (Distinct). Set the Marks type to Bar, click Show Mark Labels to display the counts, and edit the title and axis labels.
6. Click New Dashboard, rename it as "Employee Performance Dashboard", and set the size to Automatic.
7. Drag the "Performance Trend" sheet and the "Department Distribution" sheet from the Sheets list into the dashboard area. Show the dashboard title from the Dashboard menu.
8. Interactive filter: on the Performance Trend sheet, drag Department to Filters, select all departments, and choose Show Filter. In the filter's drop-down, choose Apply to Worksheets > Selected Worksheets, tick both sheets and click OK. Also click the Use as Filter icon on the bar chart so that clicking a department bar filters the line chart.
9. Test the dashboard by selecting different departments and check that both charts update. Take a screenshot of the final dashboard.



Output 4: Tableau Employee Performance Dashboard

Observation: The dashboard shows the performance trend (line chart) and the department-wise employee count (bar chart) side by side. Using the Department filter, or by clicking a bar, the user can view the data of a single department such as HR, Sales or Marketing and see how its performance scores change with years of service. This makes it easy to compare departments interactively.

PROCEDURE

1. Create vectors and a data frame (employee) in R containing Employee ID, Department, Years of Service and Performance Score.
2. Sort the data by Years of Service and create the line chart using the `plot()` function with `type = "b"`. Add the title, axis labels and legend.
3. Find the number of employees in each department using the `table()` function and create the bar chart using `barplot()`. Label the bars with the counts using `text()`.
4. Create the scatter plot of Years of Service and Performance Score using `plot()` and calculate the correlation using the `cor()` function.
5. Import the Employee Performance dataset into Tableau and convert Employee ID and Years of Service to dimensions.
6. Create the line chart (Years of Service vs Performance Score) and the bar chart (Department vs count of employees) in separate worksheets.
7. Combine both worksheets in a Tableau dashboard, add the Department filter and test the interactivity.

RESULT

Thus, the employee performance data was successfully visualized using a line chart, a bar chart, a scatter plot and a Tableau interactive dashboard. The scatter plot was used to examine

the relationship between years of service and performance score, and it showed a weak negative correlation ($r = -0.1688$) in the given dataset.